

Quantum Speed-Up of Estimating Partition Functions

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Related Articles

- Szegedy, *Spectra of Quantized Walks and a $\sqrt{\delta\epsilon}$ rule*, 2004
- Magniez, Nayak, Roland, Santha, *Search via Quantum Walk*, 2006
- Santha, *Quantum Walk Based Search Algorithms*, 2008 (overview article)
- Somma, Boixo, Barnum, Knill, *Quantum Simulations of Classical Annealing Processes*, 2008
- W., Abeyesinghe, *Speed-Up via Quantum Sampling*, 2008

Estimating Partition Functions

- let Ω be a set whose elements x correspond the states of some physical system
- let $E : \Omega \rightarrow \mathbb{R}$ denote the energy function, assigning each state x its energy $E(x)$
- given the desired (inverse) temperature β , the task is to estimate the partition function $Z(\beta)$

$$Z(\beta) = \sum_{x \in \Omega} e^{-\beta E(x)}$$

Computational Complexity

- the problem of estimating $Z(\beta)$ with high degree of accuracy for general $E : \Omega \rightarrow \mathbb{R}$ and high β is #P-hard
- $\implies$ It is unlikely that there are efficient (classical and quantum) algorithms for this task

FPRAS

- we consider fully polynomial randomized approximation schemes (FPRAS)
- a FPRAS
 - outputs a random number $\hat{Z}$ satisfying

$$\Pr \left((1 - \epsilon)Z(\beta) \leq \hat{Z} \leq (1 + \epsilon)Z(\beta) \right) \geq 3/4$$

where $\epsilon \in (0, 1)$ determines the desired accuracy

- runs in time polynomial in $\log(|\Omega|)$ and $1/\epsilon$

Simulated Annealing

- choose a cooling schedule $\beta_0 < \beta_1 < \dots < \beta_\ell$ with $\beta_0 = 0$ and $\beta_\ell = \beta$
- express the desired quantity as a telescoping product

$$Z(\beta) = \frac{Z(\beta_\ell)}{Z(\beta_{\ell-1})} \cdot \frac{Z(\beta_{\ell-1})}{Z(\beta_{\ell-2})} \dots \frac{Z(\beta_1)}{Z(\beta_0)} \cdot Z(\beta_0)$$

- observe that $Z(\beta_0)$ is trivial since $Z(\beta_0) = |\Omega|$
- $\Rightarrow$ estimate the ratios $\alpha_i := Z(\beta_{i+1})/Z(\beta_i)$

Estimation via Boltzmann Sampling I

- denote by $\pi_i = (\pi_i(x) : x \in \Omega)$ the Boltzmann distribution at inverse temperature β_i

$$\pi_i(x) = \frac{e^{-\beta_i E(x)}}{Z(\beta_i)}$$

- assume we can sample from π_i
- assume we can find a short cooling schedule so that each ratio $\alpha_i := Z(\beta_{i+1})/Z(\beta_i)$ is bounded from below by a constant, say $1/2$
- $\Rightarrow$ it suffices to take a small number of samples $\sim \pi_i$ to estimate α_i

Estimation via Boltzmann Sampling II

- let $X_i \sim \pi_i$
- let $\Delta\beta_i = \beta_{i+1} - \beta_i$
- define a new random variable

$$Y_i = e^{-\Delta\beta_i E(X_i)}$$

- the expected value $\mathbb{E}(Y_i)$ is equal to

$$\sum_{x \in \Omega} \pi_i(x) e^{-\Delta\beta_i E(x)} = \sum_{x \in \Omega} \frac{e^{-\beta_i E(x)}}{Z(\beta_i)} e^{(-\beta_{i+1} + \beta_i) E(x)} = \alpha_i$$

- $\Rightarrow Y_i$ is an unbiased estimator for α_i

Estimation via Boltzmann Sampling III

- draw $O(\ell/\epsilon^2)$ samples of X_i and compute the mean $\bar{Y}_i$
- $\Rightarrow$ the random variable $\bar{Y} = \bar{Y}_0 \bar{Y}_1 \cdots \bar{Y}_\ell$ satisfies

$$\Pr \left((1 - \epsilon)\alpha < \bar{Y} < (1 + \epsilon)\alpha \right) \geq 7/8$$

where $\alpha = \alpha_0 \alpha_1 \cdots \alpha_\ell$

- $\Rightarrow$ the total number of samples is

$$O\left(\ell^2/\epsilon^2\right)$$

Sampling with Markov Chains

- in general, we are not able to sample directly from π_i
- for some $E : \Omega \rightarrow \mathbb{R}$, we can construct a Markov chain P_i such that
 - its stationary distribution is equal to π_i
 - its spectral gap δ is large
- $\Rightarrow$ we simulate

$$\tilde{O}(1/\delta)$$

steps of P_i to obtain one sample from $\tilde{\pi}_i$, which is sufficiently close to π_i

Quantum Speed-Up

- classical complexity

$$\tilde{O}\left(\ell^2 / (\delta \epsilon^2)\right)$$

- quantum complexity

$$\tilde{O}\left(\ell^2 / (\sqrt{\delta} \epsilon)\right)$$

- $1/\delta \rightarrow 1/\sqrt{\delta}$ is due to the quadratic relation between the spectral gaps of P_i and the phase gaps of the corresponding quantum walks $W(P_i)$
- $\epsilon^2 \rightarrow \epsilon$ is due to quantum estimation of expected values of observables

Classical Walk

- P_i is a time-reversible Markov chain

$$P_i = \left(p_i(x, y) \right)_{x, y \in \Omega}$$

with stationary distribution

$$\pi_i = \left(\pi_i(x) \right)_{x \in \Omega}$$

- π_i is the unique left eigenvector of P_i with eigenvalue 1

$$\pi_i P_i = \pi_i$$

Spectral Gap

- let $1 = \lambda_0 > \lambda_1 > \dots > \lambda_{|\Omega|-1} > 0$ be the spectrum of P_i

- the spectral gap

$$\delta := 1 - \lambda_1$$

determines how fast π_i is approached from any initial state

Quantum Update

- a quantum update for P_i is any unitary U_i acting on $\mathbb{C}^{|\Omega|} \otimes \mathbb{C}^{|\Omega|}$ with

$$U_i(|x\rangle \otimes |0\rangle) = |x\rangle \otimes \sum_{y \in \Omega} \sqrt{p_i(x, y)} |y\rangle$$

for some fixed state $0 \in \Omega$

- classical complexity: the number of times we apply P_i
- quantum complexity: the number of times we apply U_i or $U_i^\dagger$

Quantum Walk

- the quantum walk $W(P_i)$ is realized by applying U_i twice and $U_i^\dagger$ twice (and also some other simple unitaries)
- the unique eigenvector of $W(P_i)$ with eigenvalue 1 is

$$|\pi_i\rangle = \sum_{x \in \Omega} \sqrt{\pi_i(x)} |x\rangle$$

- $|\pi_i\rangle$ is a coherent version of the limiting distribution π_i

Quadratic Relation

- the phase gap Δ of $W(P_i)$ is

$$\min\{|\varphi| : e^{i\varphi} \text{ is an eigenvalue of } W(P_i), e^{i\varphi} \neq 1\}$$

- the quadratic relation between the phase and spectral gaps

$$\Delta \geq \sqrt{\delta}$$

is at the heart of quantum speed-ups of many search problems

Structure of the Quantum Algorithm

$$|\pi_0\rangle \longrightarrow \boxed{\tilde{\alpha}_0} \longrightarrow$$

$$|\pi_0\rangle \longrightarrow |\tilde{\pi}_1\rangle \longrightarrow \boxed{\tilde{\alpha}_1} \longrightarrow$$

$$|\pi_0\rangle \longrightarrow |\tilde{\pi}_1\rangle \longrightarrow \cdots \longrightarrow |\tilde{\pi}_{\ell-1}\rangle \longrightarrow \boxed{\tilde{\alpha}_{\ell-1}} \longrightarrow$$

Preparation of Quantum Samples I

- the fact that the ratios α_i are bounded from below by $1/2$ implies

$$|\langle \pi_i | \pi_{i+1} \rangle|^2 \geq \frac{1}{2}$$

- $\Rightarrow$ we can drive $|\pi_i\rangle$ to $|\tilde{\pi}_{i+1}\rangle$ by applying $W(P_i)$ and $W(P_{i+1})$

$$\tilde{O}(1)$$

times

- this is based on Grover's $\frac{\pi}{3}$ -fixed point search

Preparation of Quantum Samples II

- starting from $|\pi_0\rangle$ (uniform superposition), we can prepare any $|\tilde{\pi}_i\rangle$ by

$$|\pi_0\rangle \rightarrow |\tilde{\pi}_1\rangle \rightarrow \cdots \rightarrow |\tilde{\pi}_i\rangle \rightarrow \cdots \rightarrow |\tilde{\pi}_{\ell-1}\rangle$$

- $\Rightarrow$ it suffices to apply operators in $\{W(P_k) : k = 0, \dots, i\}$

$$\tilde{O}\left(\frac{i}{\sqrt{\delta}}\right)$$

times

- $\Rightarrow$ the complexity of preparing $|\tilde{\pi}_0\rangle \otimes |\tilde{\pi}_1\rangle \otimes \cdots \otimes |\tilde{\pi}_{\ell-1}\rangle$ is

$$\tilde{O}\left(\frac{\ell^2}{\sqrt{\delta}}\right)$$

Quantum Estimation of the Ratios I

- let A_i be the observable

$$A_i = \sum_{x \in \Omega} e^{-\Delta\beta_i E(x)} |x\rangle\langle x|$$

- we have

$$\langle \pi_i | A_i | \pi_i \rangle = \alpha_i$$

where

$$|\pi_i\rangle = \sum_{x \in \Omega} \sqrt{\pi_i(x)} |x\rangle$$

Quantum Estimation of the Ratios II

• let

$$V_i := \sum_{x \in \Omega} |x\rangle\langle x| \otimes \begin{pmatrix} \sqrt{e^{-\Delta\beta_i E(x)}} & \sqrt{1 - e^{-\Delta\beta_i E(x)}} \\ -\sqrt{1 - e^{-\Delta\beta_i E(x)}} & \sqrt{e^{-\Delta\beta_i E(x)}} \end{pmatrix}$$

• let

$$|\psi_i\rangle = V_i \left(|\pi_i\rangle \otimes |0\rangle \right) \quad \text{and} \quad \Gamma := I \otimes |0\rangle\langle 0|$$

• $\Rightarrow$ we have

$$\begin{aligned} \langle \psi_i | \Gamma | \psi_i \rangle &= \langle \pi_i | A_i | \pi_i \rangle \\ &= \alpha_i \end{aligned}$$

Quantum Estimation of the Ratios III

- we obtain a random variable Q_i with

$$\Pr \left(\left(1 - \frac{\epsilon}{2\ell}\right) \alpha_i \leq Q_i \leq \left(1 + \frac{\epsilon}{2\ell}\right) \alpha_i \right) \geq 1 - \frac{1}{8\ell}$$

by applying quantum phase estimation to

$$G_i = (2|\psi_i\rangle\langle\psi_i| - I)(2\Gamma - I) \quad \text{and} \quad |\psi_i\rangle = |\pi_i\rangle|0\rangle$$

- to achieve the desired accuracy, we have to apply G_i

$$\tilde{O} \left(\frac{\ell}{\epsilon} \right)$$

Quantum Estimation of the Ratios IV

- consider the random variable $Q_0 Q_1 \cdots Q_{\ell-1}$

- $\Rightarrow$ we have

$$\Pr \left((1 - \epsilon)\alpha \leq Q \leq (1 + \epsilon)\alpha \right) \geq \frac{7}{8}$$

- the success probability decreases to $\geq 3/4$ due to imperfections

- to obtain the desired accuracy, it suffices to apply operators in $\{W(P_k) : k = 0, \dots, \ell - 1\}$

$$O\left(\frac{\ell^2}{\sqrt{\delta}\epsilon}\right)$$

times

Summary: Quantum Speed-Up

- classical complexity

$$\tilde{O}\left(\ell^2 / (\delta \epsilon^2)\right)$$

- quantum complexity

$$\tilde{O}\left(\ell^2 / (\sqrt{\delta} \epsilon)\right)$$

- $1/\delta \rightarrow 1/\sqrt{\delta}$ is due to the quadratic relation between the spectral gaps of P_i and the phase gaps of the corresponding quantum walks $W(P_i)$
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Future Research

- improving upon Chebyshev sampling on a quantum computer?
- quantum speed-up of
 - estimating permanents of matrices with non-negative entries?
 - quantum speed-up of estimating the volume of a convex polytope?
 - quantum speed-up of other classical approximation algorithms?
- estimating partition functions of **quantum** Hamiltonians